

Venkata Saratchandra Patnaik Markonda Patnaikuni

+1 6023018122 | smarkond@asu.edu | <https://www.linkedin.com/in/saratchandra-92015>

EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Project Development & Coordination: Budget tracking, cross-functional collaboration, vendor/SME coordination, regulatory documentation support

Analytical Tools: Python, R, Excel modeling, SQL, data-driven decision-making

Databases: MySQL, PostgreSQL, Google Firebase, MariaDB

Cloud & DevOps: AWS (S3, EC2, Lambda, EKS, Greengrass), Docker, Kubernetes, Linux, Shell scripting. Project Management

Tools: Jira, GitHub, ArgoCD, Jenkins, Grafana, Microsoft Project (familiarity)

Research & Analysis: Market and systems research, risk identification, performance optimization, documentation

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- Integrated Stream Deck client devices with the media dashboard to control live broadcast operations, including Go Live functions and ad playback between broadcasts, enabling real-time control and improving operational efficiency.
- Designed onboarding solutions for content creators, optimized video ingestion and distribution, and used Terraform-based automation for customer onboarding, reducing manual setup time by 90%.
- Managed 15+ containerized microservices on AWS EKS with Docker/Kubernetes, deployed GitOps-based CI/CD pipelines via ArgoCD, and wrote Python scripts for API integration and system verification, reducing failures by 95%.

Web Development Intern: Blueplanet Solutions Inc., India

April 2021 - June 2021

- Optimized MySQL database using stored procedures for 'Campus Club', improving query speed by 50% and enabling faster data access for over 1,000+ student profiles.
- Enhanced a responsive student search UI with HTML, CSS, and JavaScript, reducing search time by 60% and improving user experience.
- Analyzed web server logs to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Underground Cable Fault Detection (IoT and Embedded Systems)

- Designed and simulated a fault detection system for underground cables using Proteus and NodeMCU ESP8266, integrating voltage sensing and fault localization.
- Developed real-time monitoring with an LCD and remote notifications via the Blynk app, enabling rapid fault detection.
- Applied principles of Ohm's Law, circuit analysis, and embedded programming to automate diagnostics.
- Simulated and validated the system in Proteus, minimizing deployment errors and reducing development time.

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Built an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged a complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda; automated deployments via shell scripts and passed over 100+ real-time autograder test cases.